Protein consumption and alternative proteins in India and Pakistan

Muhammad Umar Zafar

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AI-assisted research draft. Human scholarly review is pending. This report brings together the comparative brief, food and nutrition data, consumer-study extraction, supply benchmark and cultivated-chicken work. It distinguishes source observations, analyst calculations, documentary interpretation and unresolved questions. The accompanying edition presents the same evidence in a shorter visual sequence.

The comparison is informative but an adoption baseline is still missing

India and Pakistan differ in the food available nationally, the quantities recorded by household surveys and the institutions relevant to new food products. These differences identify questions for a comparative study. They do not establish which country will adopt alternative proteins faster, how many consumers will buy them, or whether purchases will displace animal foods.

The principal findings are:

Pakistan's poultry availability was 8.72 kg per person per year, compared with 3.52 in India, in 2023. These are FAOSTAT food-balance quantities available for human consumption.

India's pulse availability was 15.39 kg per person per year, compared with 6.53 in Pakistan in the same year. Familiar plant foods therefore belong in a study of competing meals; pulse supply does not measure demand for a manufactured meat analogue.

The State of Food Security and Nutrition in the World (SOFI) estimates that 40.4 percent of India's population and 60.3 percent of Pakistan's could not afford a healthy diet in 2024. Economic access is a separate outcome from measured dietary intake.

The selected Joint Child Malnutrition Estimates (JME) country model estimates under-five stunting at 32.9 percent in India and 33.6 percent in Pakistan for 2024. These modelled estimates cannot be added to unaffordability or undernourishment counts.

India's official household-based protein estimates can be compared with adult requirement scenarios only after retaining the published consumer-unit denominator. The consumer unit is based on energy requirements, so this comparison cannot establish protein adequacy or population deficiency.

Pakistan's previously displayed complete protein total is withdrawn. Named physical conversions now describe a partial food basket, with unresolved foods and denominator reconciliation left visible. No national adequacy estimate or rural and urban model ranking survives that repair.

The supply benchmark predicts national protein availability under exploratory models. Its held-out interval results differ sharply between the country series, and its forecast intervals do not supply probabilities of alternative-protein adoption.

The consumer-study register contains 9 studies and 29 extracted findings. These are heterogeneous published results, not respondent microdata or a bilateral acceptance estimate.

India has documented research support and food-regulatory instruments; Pakistan has halal institutions and a Punjab instrument containing a general novel-food provision. Neither a programme nor a rule establishes approval or sales of a named cultivated product.

The halal work identifies product, process and institutional conditions. A religious position, a halal certificate and food authorization must each be established for the case being assessed.

The main unresolved empirical requirement is a compatible observation of consumer behaviour for a specified product, place and period. The current record does not provide that baseline. The most useful next work is to obtain eligible respondent data, verify actual product availability and test questions about price, meal use and repeat purchase before making an adoption forecast.

A common meal makes the comparative question testable

The India–Pakistan brief asks when a new protein product earns a recurring place in a household meal. It distinguishes willingness to try from purchase, habitual use and the food displaced. That distinction remains the report's organizing question. National food supply describes the context in which people choose; it does not observe their response to a new product. The brief is a secondary synthesis, with no project consumer experiment or interview evidence.

The primary proposed comparison is a plant-based chicken product, a conventional chicken option and a familiar pulse-based meal. The options need specified serving sizes, prices, preparation demands and product descriptions. A cultivated-meat vignette would be analysed separately because its animal origin and production process introduce different questions. Precision-fermented ingredients and fungal biomass also require their own descriptions. A response to one category cannot stand for acceptance of all alternative proteins.

The brief's argument has an economic rival: price, taste and availability may account for differences that appear cultural. Comparable purchase conditions might reveal that differences within a country exceed those between selected groups in the two countries. Religious affiliation, urban residence or an aggregate food quantity cannot settle that question. The India–Pakistan research brief proposes a comparison that can distinguish these rival explanations.

The foundational plan established an India–Pakistan research lane alongside support for forecast-question design. Its early policy-attention hypothesis was a proposal to investigate, not an empirical result. The later halal work expands the documented scope without cancelling the original comparative question. This report connects the workstreams through product-specific evidence while retaining their different methods.

A forecast question needs a named product category, geography, outcome, denominator, deadline and resolution source. Retail availability, repeated purchasing and animal-food displacement are different events. Where no compatible baseline or resolution source exists, the question remains a monitoring or data-acquisition task. Repeated model answers do not supply the missing observation, and disagreement between language models is not a calibrated probability interval.

The drivers are hypotheses about decisions

Affordability concerns what a household can buy relative to familiar meals. The relevant comparison is a delivered serving at an actual outlet, including preparation and access costs where measured. A laboratory cost estimate or premium retail offer cannot establish a typical household's choice. The household expenditure gradients in this report motivate price-sensitive questions without identifying an elasticity for alternative proteins.

Dietary practice concerns ingredients, meal roles, preparation and the people who buy and cook. A product resembling meat may appeal to someone seeking a substitute and repel someone avoiding meat-like foods. Both possibilities fit the present evidence. A questionnaire should therefore ask what the proposed product would replace and whether it fits an ordinary meal, rather than treating resemblance as a universal advantage.

Social legitimacy concerns whose approval matters and how a choice is understood in the household or community. The report's culture matrix contains descriptive context, not a causal ranking of religions or nationalities. Region, income, caste, gender, access and dietary practice can overlap. Their independent contributions require a design that measures those relationships instead of assigning them from a national label.

Certification concerns the credibility and scope of an assurance for a named product. A consumer's belief, a religious authority's conditions and a certifier's decision are separate observations. For cultivated meat, donor history and culture inputs may matter as much as the final label. A generic assertion that a product is animal-free cannot resolve questions about every earlier production stage.

Public policy concerns permission to sell, research support, standards, disclosure and the institutions responsible for those decisions. The plausible connection to consumption is through products, prices, access or trust. The current documentary evidence records instruments and events; it does not identify their effects on purchases. A funded project remains distinct from a commercial product reaching a household.

The proposed sequence from acceptability to trial, purchase and habit is an analytical framework. It is not an estimated causal model. Trial could replace meat, replace a familiar plant dish or add an occasional novelty. Repeated purchases and the displaced food would need to be observed before the study could assess substitution.

Different data describe different parts of the food system

The protein dashboard supplies the national, regional, dietary, nutrition, company, policy and cultural tables used here. Its repaired datasets preserve source identifiers, units, vintages and row-level evidence tiers. A source ledger describes the publication or data source; a raw manifest records downloaded bytes and retrieval dates. A successful download establishes retrieval, while a source locator and compatible interpretation establish what a row can support.

The report uses official food balances for national availability, household surveys for their stated household-based measures and nutrition indicators for their own populations and estimators. Analyst calculations carry a derived tag. A calculation based on an official table does not inherit the table's observational status. Missing values remain blank, and the edition avoids drawing lines through them as if they were observed zeros.

Every displayed quantitative claim has a ledger entry specifying a dataset, exact filter, operation, unit, source identifiers, locator and rounding rule. The verification program recomputes values from the selected rows. It also checks visible numbers, chart data, source joins, package coverage, review evidence, browser receipts, prose, public links, publication files, the library build and export text. Deliberate corruptions tested these checks before the edition and report were built. The verification record describes what those checks actually establish; it does not confer scholarly approval.

AI agents collected and processed sources under the author's direction, assembled the tables and drafted this synthesis. The source records retain access limits and unresolved interpretations. The agents did not conduct a consumer survey, obtain project interview evidence, certify a product or establish a causal effect. Human scholarly review remains pending.

The earlier data assessment is a dated scoping document. It reviewed a selected set of files and identified useful food-balance material, gaps in data access and possible policy distinctions. Subsequent source repair changes how some of its statements can be used. Its description of an older unopened halal register does not describe the separately audited paper register, and its Pakistan policy-absence language is superseded by the Punjab evidence discussed below.

The assessment is preserved rather than rewritten retrospectively. Its contribution is to the history of the research question and the identification of inputs. Current numerical findings come from the repaired tables and claim ledger. A historical assessment, a later source audit and independent human verification have different evidential standing.

The approved design preview supplied the editorial visual direction: readable prose, separate country colours, source lines and direct access to tables. It was a precursor to the full edition. Its old figures and browser receipts are historical, so they cannot validate the repaired page. The final edition reads the current processed data and binds its browser checks to the actual HTML bytes.

Regional maps retain the selected boundary source and its original metadata. Jammu and Kashmir, Ladakh, Azad Kashmir and Gilgit-Baltistan are hatched without a data colour; values remain available in the corresponding source tables where present. This treatment prevents the colour scale from implying a territorial determination. It does not reconcile differences between survey geography and the boundary edition. Narrow screens use ranked displays where a map would obscure the evidence.

Pakistan has more poultry availability while India has more pulses

FAOSTAT records 8.72 kg of poultry available per person per year in Pakistan and 3.52 in India in 2023. For pulses, the order reverses: 15.39 kg in India and 6.53 in Pakistan. Milk excluding butter is 87.43 kg in India and 129.71 in Pakistan. The source is the saved FAOSTAT food-balance release, with its flags and commodity definitions retained. FAOSTAT food balances

Food balance sheets estimate food available for human consumption within a national accounting system. They do not measure an individual's ingestion, household distribution or retail sales of alternative proteins. The series can compare recorded availability under the source's definitions. It cannot identify who consumed the food, how much was wasted after supply entered households or which products competed at a particular meal.

The pulse comparison matters because familiar plant foods are already part of ordinary food systems. Large pulse availability may be compatible with interest in a meat analogue, indifference to it or preference for a familiar dish. None of those responses follows from supply alone. A study that compares an analogue only with meat would miss a relevant existing option for some consumers.

The OECD–FAO Agricultural Outlook uses a separate method and weight convention. Its poultry projection is shown apart from the historical food-balance series, with its vintage and retail-weight unit. Joining the projected value directly to the FAOSTAT endpoint would create an apparent change partly from incompatible definitions. The report treats the outlook as a conditional projection from its source rather than an extension of measured household consumption. OECD–FAO Agricultural Outlook

For 2034, that outlook projects poultry consumption at 2.30 kg per person per year in India and 5.70 in Pakistan, in retail weight. The source release covers 2025–2034. The numerical difference from the food-balance endpoint is not an estimated decline under a common measurement system.

Regional food frequency and household quantities answer different questions

India's National Family Health Survey (NFHS-5) measures reported food frequency among specified respondent groups. Its fish, chicken or meat category combines foods and records frequency rather than portion size. The regional panels retain the sex, age group, residence, fieldwork period and tabulation base applicable to each row. A weekly-frequency percentage cannot be converted to kilograms or grams of protein without information the table does not provide. National Family Health Survey

Pakistan's Household Integrated Economic Survey, HIES, supplies household-based quantities and expenditure measures. The revised quantity workbook reports monthly per-capita quantities, so calling them household totals would change their meaning. Its regional chicken rows can describe differences within that survey's design. They cannot be placed on the same colour scale as India's reported weekly food frequency and interpreted as a bilateral dietary ranking. HIES revised quantity table

The Pakistan National Nutrition Survey contributes separate nutrition context. Its population and indicators differ from HIES food quantities. The maps and tables retain geographic coverage gaps and distinguish the former merged areas where the source requires it. A missing join to the chosen boundary file is an unresolved mapping problem, not a zero observation.

Residence differences in these tables are descriptive. Urban and rural respondents or households differ in many measured and unmeasured circumstances. Neither the map nor the residence toggle identifies an effect of urbanisation. The useful follow-up is to examine access, prices, expenditure and meal practices in specified locations before selecting study sites.

Food budgets and protein shares show different dietary structures

Pakistan's chicken quantity rises across the selected expenditure quintiles in the revised HIES table. This is a per-capita quantity gradient within one source, not a measured preference for an alternative product. Household size also differs across these groups, which makes denominator discipline necessary. A household total and a per-capita quantity can order groups differently without either describing the same estimand.

The lowest quintile records 0.17 kg of chicken per person per month, compared with 0.65 in the highest. The accompanying household-size context falls from 7.56 to 4.65 persons across those groups. The quantities come from the revised HIES 2024–25 workbook, with the precise cells recorded in the claims ledger. The exact analytical base for that revised table is not established by the report's broader household-count context.

India's Household Consumption Expenditure Survey, HCES, converts reported household food into nutrient estimates and reports protein contributions by food group. Cereals, pulses, dairy and the combined eggs, fish and meat group each have a distinct role. The combined animal-food group in that table excludes dairy. Its share cannot be described as all animal-source protein. Nutritional Intake in India

The Indian fractile labels rank the population by monthly per-capita consumption expenditure (MPCE), from lower to higher spending. Rural and urban populations are ranked separately. The labels describe positions in those expenditure distributions; they are not percentages below a protein requirement or shares of the food budget.

These tables broaden the comparative question beyond replacing meat. A product might compete with a cereal accompaniment, a pulse dish, dairy or a convenience meal. The report cannot determine that competition from ingredient composition. It requires asking consumers about a specified occasion and recording the food actually replaced.

Expenditure measures need equal care. A food-budget share is a proportion of food expenditure, not total household expenditure, energy or protein. A quantity gradient is not a price elasticity. The source rows retain those distinctions so that apparent agreement between indicators does not create an unsupported common scale.

The Indian benchmark is a scenario and the Pakistan total is missing

India's published consumer-unit estimates are the appropriate denominator for the report's adult reference scenarios. A consumer unit expresses energy needs relative to a sedentary adult man. It is not a protein-equivalent person. Comparing a household group's mean protein estimate with an adult requirement therefore supplies a descriptive scenario, with an explicit mismatch that prevents an adequacy conclusion.

The all-India estimates are 66.6 g per consumer unit per day in rural areas and 69.9 in urban areas. HCES fieldwork ran from August 2023 through July 2024; these months describe the survey period, not an individual respondent's observation interval. The nutrient table's analytical base remains distinct from the survey's overall household sample.

The Estimated Average Requirement, EAR, is the primary reference; the Recommended Dietary Allowance, RDA, is retained as a labelled sensitivity. They answer different questions about requirements and cannot be swapped merely to produce a larger apparent gap. The report also retains alternative body-weight assumptions. None of these comparisons observes the distribution of usual individual intake needed for an estimate of population inadequacy.

At a reference weight of 60 kg, the EAR scenario uses 0.66 g per kg per day, or 39.60 g per day. The RDA sensitivity uses 0.83 g per kg per day, or 49.80 g per day. These calculations are tagged derived even though their requirement coefficients come from reference sources. They are adult scenarios without a

protein-equivalent household denominator.

A group mean above a reference can coexist with individuals below it. A group mean below a reference does not tell how many individuals are below it. Counting expenditure groups is especially misleading when those groups differ in population or household composition. The former unweighted fraction of groups below a benchmark has therefore been removed rather than described as a population deficiency rate.

Pakistan's former full-diet estimate used a pooled-basket cost calibration that did not establish complete protein intake. The current model withdraws that total and preserves it only as historical output. The replacement converts selected reported food quantities through named food-composition factors and edible-portion assumptions. It reports a matched-food subtotal and a separate subtotal for chicken, beef, mutton, eggs and fish.

The composition evidence comes from the Pakistan Food Composition Table, Food Composition Table for Bangladesh, Indian Food Composition Tables and USDA SR Legacy release. Pakistan-table food entries anchor the local matches; the Bangladesh Rohu entry supplies distinct edible-portion assumptions, while Indian and USDA entries supply explicitly named food-form alternatives. The INFOODS density database supports volume-to-mass assumptions. USDA documentation defines the edible-portion basis; its portion weights do not establish Pakistani egg-size frequencies. The model also treats tea-transfer and product-mixture choices as conditional assumptions. These sources provide composition and method inputs, not measured Pakistani intake.

For each named endpoint, the scenario-factor table identifies the conditional food conversions. Its factor identifiers join the source-factor catalogue, which gives original PDF pages, food codes, units and source-file hashes. The claims ledger preserves that join and the complete composition-source list alongside the household quantity cells. The method guide explains how to follow this record without treating an assumed food match as an observed product mix.

The national matched-food subtotal is 39.27 to 40.61 g per person per day. The separate chicken, beef, mutton, eggs and fish subtotal is 3.30 to 4.64 g per person per day. The low method is `bd_cuts_rohu_us_small`; the high method is `edible_flesh_us_jumbo`. Their differences concern edible portions, fish matching and egg-size assumptions, rather than sampled uncertainty.

Of 38 food rows, 27 receive a conditional conversion; 0 are imputed. The separate animal-food subtotal excludes dairy and other animal-source foods. Neither subtotal is called total protein intake, and no quintile ordering is inferred from these incomplete sums.

The method range expresses the extrema among named conditional conversions. It is not a sampling confidence interval or a complete uncertainty bound. Unresolved foods, prepared-food coverage, food-form mixtures and quantity reconciliation remain outside the subtotal. No imputation is used to silently complete the diet. The range therefore remains separate from the Indian consumer-unit panel and from the adult EAR/RDA scenarios.

The unresolved food quantities and local product mixtures are the next inputs needed before a complete Pakistani intake estimate can be attempted.

Affordability and child growth reveal overlapping concerns

SOFI estimates that 40.4 percent of India's population and 60.3 percent of Pakistan's could not afford a healthy diet in 2024. The measure combines an economic-access model with a specified healthy-diet cost. It does not observe every household's actual food purchases or diagnose a nutrient deficiency. The selected source vintage remains attached to the figures. The State of Food Security and Nutrition in the World

The JME country model estimates under-five stunting at 32.9 percent and 33.6 percent respectively for 2024. The country-model release and survey observations appear separately because a modelled annual estimate is not a new household measurement. The model rows retain uncertainty intervals and provenance where supplied. Joint Child Malnutrition Estimates country model

Stunting reflects accumulated growth constraints with multiple causes. It cannot be used as a protein-deficiency measure or as an addressable market for a meat substitute. Undernourishment likewise uses its own definition, population and estimation procedure. The same person can fall within several indicators, and their reference periods can differ. Adding their counts would double-count people and combine incompatible outcomes.

The report therefore presents an explicit instruction not to add these indicators. Their joint relevance is to research design: affordability, dietary quality and child growth deserve attention, but each needs an appropriate intervention question. The current evidence does not establish that a particular alternative-protein product is affordable, suitable for the affected population or effective against any of these outcomes.

Exploratory supply forecasts need better calibration evidence

The standalone benchmark models annual national protein supply using frozen historical series. Total and animal-source protein are separate series for each country. Its measured outcome is available supply, not alternative-protein purchases, consumption frequency or an adoption share. The selected forecast horizon extends to 2035, subject to the benchmark's assumptions and short observed history.

The frozen input contains 56 annual observations and the selected output contains 48 forecast records. The protocol defaults to persistence because its model-selection evidence is too short to support a more ambitious selection claim. Consequently, the selected point forecasts retain the last observed level while uncertainty grows with the horizon. A flat point path is a stated benchmark assumption, not evidence that supply will be constant.

Selected persistence forecasts for 2035, grams per person per day. The lower and upper bounds are the benchmark's nominal 80 percent interval, with the coverage limitations below.

Country	Protein supply	Point	Lower	Upper
India	Animal-source	20.01	16.67	23.35
India	Total	74.57	68.14	81.00
Pakistan	Animal-source	32.55	28.50	36.60
Pakistan	Total	73.39	65.23	81.55

Source: Protein-supply benchmark. All forecasts are derived.

At the nominal 80 percent interval level, the selected total-supply interval covers 1 of 6 held-out Indian cells and 6 of 6 Pakistani cells. The animal-source series has the same country-specific coverage counts: 1 of 6 for India and 6 of 6 for Pakistan. The evaluation rows and split remain attached to these results.

The held-out comparison reports very different interval coverage across countries. The report retains the actual counts and evaluation denominator alongside the nominal interval level. A small set of overlapping evaluation cells cannot demonstrate stable calibration. Broad intervals may cover observations without producing useful precision, while narrow intervals can miss them repeatedly. Neither result supports interpreting the forecast as a calibrated statement about future consumers.

The uneven held-out coverage is a reason to investigate the interval method before using these supply scenarios for decisions. Neither the calibration exercise nor its forecast contains a measured alternative-protein market share.

The missing adoption baseline cannot be supplied by relabelling this model. A future adoption study would need observations of the relevant product and outcome, a compatible denominator and a resolution source. Until those inputs exist, the supply benchmark remains a separate research product.

The package also contains statistical diagnostics, a hypothesised causal graph and a policy calculator with assumed pilot budgets. These describe proposed relationships or analyst-selected scenarios. Installed causal-analysis software does not identify an effect, and a calculator’s arithmetic does not validate its national parameters. The evidence index supports source retrieval without generating a new consumer observation.

Industry records establish events rather than a market total

The company register distinguishes a business, a product category, a dated event and its evidence. A directory listing is not proof that a factory operates at its stated capacity. A company website can document a claim or offer without independently confirming sales. Closures and exclusions remain visible so that an earlier optimistic record is not silently carried forward as current operation.

UnCrave’s reported closure is recorded as a dated event. The announcement date does not establish the exact date on which every product ceased to be available. The Phyx company lead is recorded as an India lead rather than a Pakistan company. The food/feed rule excludes feed-only activity from human-food company totals and retains adjacent or mixed-use businesses with explicit category limits. A traditional soy-food or plant-milk business does not become a meat-alternative producer merely because its inputs are plant-derived.

Pakistan’s identified retail material is similarly bounded. A catalogue or import-to-order offer demonstrates that a seller presented a product under particular terms. It does not prove current stock, completed sales, nationwide availability or repeat purchasing. Dated secondary reporting on a business remains reporting from that date unless a newer operational source is recovered.

Pulse production provides potential feedstock context, with crop definitions and source flags retained. Whole-crop tonnage can have several end uses and cannot be converted into alternative-meat output, isolate capacity or sales without additional process and allocation evidence. Aggregates and constituent crops also overlap, so they are not added as independent supplies.

The bounded public-source search recovered newer industry, consumer-awareness, export and global-market material. It did not establish a compatible newer estimate of India's domestic alternative-protein market. This is a narrower conclusion than saying no newer GFI information exists. Historical national scenarios, global sales and export opportunities remain different quantities.

Published consumer studies leave the bilateral comparison open

The survey extraction contains 9 study records and 29 findings. These studies differ in country, product, recruitment, wording and outcome. They are not all studies of India and Pakistan. The extraction contains a Pakistan sample descriptor but no India finding, so its row count cannot be presented as a bilateral consumer dataset.

Sample size describes the reported study or analytical group, not the representativeness of its recruitment. A large online convenience sample can still omit people without internet access or people unwilling to answer in the questionnaire's language. A national label in a paper therefore needs the actual sampling description before the result is interpreted as national demand.

The report retains the studies separately in a country-and-measure table. It does not pool percentages, add overlapping respondents or estimate a common acceptance rate. A descriptor, a willingness category and a hypothetical choice result have different denominators. Apparent agreement between them does not create a comparable outcome.

Study and sample record
Chriki et al. (2024). Attitudes toward artificial meat in Arab countries. Geography: Arab countries, pooled. Topic: Artificial-meat attitudes. Reported full study sample: 1025. Source
Maqsood et al. (2025). Are Emirati consumers in United Arab Emirates open to alternative proteins?. Geography: UAE nationals. Topic: Alternative-protein preferences. Reported full study sample: 1666. Source
Dean et al. (2023). Should I Really Pay a Premium for This?. Geography: Multinational sample including Pakistan. Topic: Willingness to pay a premium. Reported full study sample: 4488. Source
Alptekin et al. (2026). Consumer acceptance of cultured meat and its determinants among Turkish adults. Geography: Turkish adults. Topic: Cultured-meat acceptance. Reported full study sample: 1169. Source
Alabdulmohsen and Al-Otaibi (2025). Exploring Novel Foods: Edible Insect Acceptance in Saudi Arabia. Geography: Saudi adults. Topic: Edible-insect acceptance. Reported full study sample: 2208. Source
Terano et al. (2023). Exploring Muslim Consumers' Acceptance of Cultured Beef Meat. Geography: Urban Muslim consumers in Selangor. Topic: Cultured-beef acceptance. Reported full study sample: 102. Source
Rakuten Insight (2025). Plant-based food alternatives, June 2025. Geography: Asian markets. Topic: Extracted findings: willingness to eat cultivated meat. Reported full study sample: 78816. Source
GFI APAC / The Good Growth Co. (2024). Decoding Demand. Geography: Selected Southeast Asian countries. Topic: Alternative-protein demand measures. Reported full study sample: 5971. Source
Lwin, Malik, Ho and Schulz (2026). Feeding the future. Geography: Singapore. Topic: Product-specific consumer measures. Reported full study sample: 2000. Source

The sample column is the source register's total study sample, not a common analytical denominator. Publication dates in titles are not fieldwork dates. Recruitment and exclusions remain in the accompanying study CSV and original sources.

The following source-verified outcomes illustrate distinct questions. The country panels, category preferences and reported prior insect consumption remain separate.

Rakuten Insight (2025). Plant-based food alternatives, June 2025

Would you eat lab-grown meat?. Fieldwork: Not established from the checked methodology panel. Recruitment: Rakuten Insight proprietary online panels.

Geography and product	Source result, percent	Reported group sample
Indonesia; Cultivated meat	31.0	8305
Malaysia; Cultivated meat	50.0	7706
Philippines; Cultivated meat	37.0	7474
Singapore; Cultivated meat	35.0	3507
Thailand; Cultivated meat	32.0	10176
Vietnam; Cultivated meat	9.0	1942

Respondents in the named country or subgroup; source-reported sample, with item nonresponse not independently recounted. Hypothetical willingness among panel respondents; no nationally representative weighting established by this check. Original source.

Maqsood et al. (2025). Are Emirati consumers in United Arab Emirates open to alternative proteins?

Preference among the alternative-protein categories presented (paraphrase). Fieldwork: January to April 2022. Recruitment: Online questionnaire among UAE nationals; young and female respondents are overrepresented.

Geography and product	Source result, percent	Reported group sample
UAE nationals; Plant-based protein	74.0	1666
UAE nationals; Cultivated meat	12.0	1666
UAE nationals; Microalgae protein	9.0	1666
UAE nationals; Insect protein	5.0	1666

Respondents in the named country or subgroup; source-reported sample, with item nonresponse not independently recounted. Category preference in this recruited sample is not purchase, willingness to pay or national adoption. Original source.

Alptekin et al. (2026). Consumer acceptance of cultured meat and its determinants among Turkish adults

Willingness to try cultured meat (abstract outcome label; full instrument wording not checked). Fieldwork: December 2023 to September 2024. Recruitment: Online cross-sectional survey of Turkish adults.

Geography and product	Source result, percent	Reported group sample
Türkiye; Cultivated meat	23.0	1169

Respondents in the named country or subgroup; source-reported sample, with item nonresponse not independently recounted. A hypothetical response among online participants, not observed tasting or purchasing. Original source.

Alabdulmohsen and Al-Otaibi (2025). Exploring Novel Foods: Edible Insect Acceptance in Saudi Arabia

Previous experience of eating edible insects (results-table outcome, paraphrase). Fieldwork: October 2024 to January 2025. Recruitment: Online snowball recruitment of Saudi adults.

Geography and product	Source result, percent	Reported group sample
Saudi Arabia: men; Edible insects	24.1	758
Saudi Arabia: women; Edible insects	7.8	1450

Respondents in the named country or subgroup; source-reported sample, with item nonresponse not independently recounted. Reported prior consumption is distinct from hypothetical willingness; non-random recruitment limits generalization. Sex-specific denominator. Original source.

The remaining extracted findings stay in the companion CSV. The approximate Arab-country willingness figure is held because the abstract supports a lower-bound description, and the Selangor paper retains conflicting acceptance figures. Summary-only outcomes whose question bases remain unresolved are also excluded from this synthesis. A verified study sample does not verify every finding attributed to that study.

Bilateral studies guide different research questions

Bryant and colleagues' 2019 paper, *A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China*, reports an Indian subgroup of 1024 respondents. Its English-language online panel is disproportionately urban, affluent and educated. The questionnaire concerns hypothetical purchases after product descriptions; it does not measure observed sales. The saved original supports the subgroup size and selection limits, but an exact fieldwork period was not located. Bryant and colleagues, Participants and Limitations. This study suggests questions about which consumers are represented before any national inference is attempted.

Arora, Brent and Jaenicke's 2020 *Is India Ready for Alt-Meat?* uses a face-to-face discrete-choice survey in Mumbai from December 2018. The English/Hindi instrument compared conventional, plant-based and cell-based meat with chana. Recruitment near railway markets and in malls or supermarkets sought varied incomes; the analysis excludes incomplete interviews. The original methodology and instrument descriptions are available in the publisher's full paper. Its local stated choices make the familiar meal alternative worth investigating, without identifying national buying behavior. No willingness-to-pay coefficient or acceptance percentage from this paper enters the current numerical synthesis.

Ahsan, Uzair and Ali's *Attitudes and Perceptions Towards Cultured Meat Among General Population in Pakistan* used social-media recruitment and an online questionnaire, with interviewer assistance where literacy prevented self-completion. The original methods place fieldwork in January to March 2021; its title page distinguishes online publication in 2021 from the 2022 journal issue. Ahsan, Uzair and Ali, Study

Design and Results (publisher access unavailable in the recorded direct-link audit; original bibliographic URL retained in the companion source CSV). A research-browser reading of the original publisher PDF confirmed these descriptions, while the direct-link audit remained blocked. Sampling limitations and inconsistent acceptance reporting prevent selecting a single percentage as Pakistan's position. The instrument still identifies safety, price and religious compatibility as questions for a better specified local study.

Dean and colleagues' multinational paper examines plant-based, fungal and cultured-muscle alternatives in separate models. Its pooled analysis does not supply a Pakistan-specific coefficient or an estimate for precision-fermented dairy. Dean and colleagues, product-specific models. The related mycoprotein paper reports a Pakistan subgroup of 649 from 2018–2019 fieldwork. The subgroup is young and nonprobability-based, and the paper retains inconsistent pooled totals. Dean and colleagues, mycoprotein methods and country sample table. Its data-request route is distinct from having respondent records suitable for reanalysis.

The Phase Two search is an access-route inventory. Its source records identify publications, repositories and possible contacts, with a separate contextual register. Discovery of a DOI, repository page or data-availability statement is not acquisition of respondent records. The reported search checks establish the organization of those routes, not eligibility for a pooled analysis.

The inventory contains 43 access-route records and 13 contextual entries. These counts describe records, not independent datasets or respondents. Overlapping papers and routes remain separate from a count of acquired data.

The next requests concern Dean's Pakistan data, the India Mendeley route and the Bryant OSF materials. Each requires confirmation of study identity, version, variables, sample definition, reuse terms and whether the available files actually contain respondent observations. A route whose title resembles a known paper remains unresolved until those checks succeed.

The planned India request concerns the Mendeley listing with a reported sample of 257. The separate listing with 350 is an additional discovery lead. Neither listing establishes an acquired or usable respondent dataset, and the larger listed sample does not replace the planned request.

Request drafts exist but remain unsent. The research should first determine what each file could answer: product understanding, stated choice at a specified price, predictors within the sampled group or observed repeat purchase. It should not request data under an adoption label when the instrument measured only hypothetical willingness.

Halal access depends on the named product and decision

The cultivated-chicken paper asks what must be established before a specified product reaches Muslim consumers through the relevant religious, certification and food-authorization processes. Its framework separates donor procurement, culture inputs, transformation, category and edibility, certification, and food authorization. These are analytical questions for organizing the evidence, not a claim that every authority reasons in the same sequence. Cultivated Chicken and Halal Market Access

The International Islamic Fiqh Academy's 2025 resolution makes its cultivated-meat position conditional on the donor, culture inputs and oversight. It excludes prohibited culture media and ingredients, requires supervision by a credible specialist authority, and assigns disclosure and verification duties. Its donor

wording distinguishes a living lawful animal from cases requiring lawful slaughter; that wording needs interpretation within the full resolution. IIFA cultivated-meat resolution. These conditions do not constitute certification of a particular manufacturer's product.

Singapore's Islamic Religious Council (MUIS) stated in 2024 that the cell source must be a halal animal, ingredients contributing to texture and composition must be halal, and the product must be clean and non-toxic. MUIS cultivated-meat guidance. That guidance concerns cultivated meat under Singapore's religious framework. It does not grant Pakistani food authorization or determine consumer willingness.

A founder cell line means cells selected and banked to start later production. The original animal sample, that banked line and the later biomass can raise different questions. A serum-free production entry does not by itself document all earlier banking stages. Washing, measured residues, biological growth and legal transformation are distinct claims. The paper leaves the evidential connections between those stages open where a product dossier is missing.

The source audit distinguishes admitted evidence, held material and sources not retrieved. Every category retains its human-review status. Classical passages, modern institutional decisions, applicant dossiers and academic interpretations have different authority and scope. An English abstract does not support a reconstruction of an unread full text, and a regulator's hosting of an applicant dossier does not make it a religious ruling.

The paper register contains 95 records: 63 verified with a note by the recorded checking process, 22 held and 10 not retrieved. All 95 retain human verification pending. The phrase verified with a note must therefore be read as a bounded source-audit status, not as independent scholarly approval.

The shorter halal brief condenses that framework for a reader who needs the main conditions and common misreadings. Its contribution is explanatory rather than a separate body of consumer data. Its Pakistan food-route statement reflects the earlier bounded search and must now be read alongside the Punjab instrument recovered for this edition.

The correction does not establish a cultivated-specific approval. It establishes that a general novel-food provision exists within the identified provincial instrument. Religious conditions, certification requirements and applicable food law still need to be matched to the product, jurisdiction and date. A later source can correct a documentary gap without resolving the whole market-access question.

The earlier cultivated-chicken comparative package assembled country dossiers, source-linked claims and a scenario-model structure. Its numerical scenario output remains blocked by a missing compatible baseline. An empty result is retained as an empty result. The existence of a tested calculation routine does not authorize invented inputs or turn a documentary framework into a national adoption estimate.

The package helped separate evidence about permission, commercial readiness and demand. Those distinctions remain useful when reviewing a named product. A complete current dossier may settle a narrow institutional question even while broader scholarly disagreement continues. Conversely, a public announcement or conditional position cannot fill missing certification, authorization or consumer evidence.

Policy instruments have distinct legal and commercial meanings

The Food Safety and Standards Authority of India (FSSAI) administers the food-regulatory instruments discussed here. India's record includes research support under Biotechnology for Economy, Environment and Employment (BioE3), vegan-food rules, the non-specified-food framework and dated labelling activity.

The parliamentary update on supported smart-protein projects is evidence of programme implementation, not evidence of food approval or repeat purchasing. The report keeps instrument dates, current document dates, proposed changes and future commencement dates separate.

Instrument and date record
India: BioE3. 2024-08-24. Cabinet approval date. Status: innovation policy approved. Source (publisher access unavailable in the recorded direct-link audit; original bibliographic URL retained in the companion source CSV)
India: DBT-BIRAC Smart Proteins call. 2025-02-22. Call announcement date; deadline recorded separately in pathway. Status: historical call closed. Source
India: Non-specified-food compendium of earlier regulations. 2022-10-14. Compendium version; underlying regulation 2017. Status: general prior approval framework. Source
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Pakistan: General requirements for halal food. 2022. Date of the named instrument, not product approval or live operations. Status: officially listed general halal standard. Source
Pakistan: Requirements for halal certification bodies. 2022. Date of the named instrument, not product approval or live operations. Status: officially listed certification body standard. Source
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India: Vegan-food import-certificate amendment. 2026-09-28; Published 2026-09-29; Commences 2027-04-01. Dated notice; publication and commencement separately recorded where established. Status: notified future commencement. Source
India: BioE3 parliamentary implementation update. 2026-03-11. PIB publication date; not each facility's commissioning date. Status: official implementation update. Source (publisher access unavailable in the recorded direct-link audit; original bibliographic URL retained in the companion source CSV)

Dates retain the source's event basis. The accompanying policy CSV separately records publication, application and commencement where established.

The dairy-analogue labelling consultation is recorded as a consultation. Later amendments have their own stated commencement dates, so publication cannot be treated as immediate application. A filing associated with Biokraft is likewise distinct from its appearance on a later status list. A procedural stage or ambiguous list entry is not enlarged into a final permission.

Pakistan's record includes the Pakistan Halal Authority and relevant standards as well as the Punjab Food Authority instrument. The latter was notified before its later gazette publication and contains a novel-food provision within a functional-food and supplement framework. The source does not establish a cultivated-specific authorization or prove that every cultivated product follows an identical route.

The correct comparison therefore concerns the documented functions of institutions and instruments. It cannot be reduced to India having policy and Pakistan having none. Nor can a count of policy rows establish regulatory effectiveness. The unresolved practical question is which documents a specified product would need and whether the responsible institutions have issued them.

Cultural context describes variation without predicting demand

The culture matrix records dimensions such as dietary identity, religion and food practice, geography, household context and relevant institutions. Each entry retains its source and date. These observations help design questions and identify populations omitted by a proposed sample. They do not yield a national willingness score.

Vegetarian self-identification differs from reported frequency of eating fish, chicken or meat. A respondent's definition can differ from a researcher's category, while frequency alone does not reveal quantities or motives. The report does not turn either measure into a market segment for a manufactured analogue. A person already avoiding meat has familiar alternatives that need to be included in the choice set.

Pakistan's urban share and halal institutions also describe different dimensions. Population geography does not measure exposure to alternative proteins, and an institutional mandate does not measure personal trust. Relevant follow-up questions would record actual shopping access, information exposure, household decision-making and product-specific assurances rather than infer them from residence or affiliation.

The strongest interpretation supported here is that a uniform national consumer does not describe the evidence. Whether particular social or religious conditions independently affect purchase remains a question for a comparable study with specified products and prices.

Review repairs change what the report can claim

The independent dashboard review identified problems in denominators, model construction, evidence tiers, policy interpretation, company scope, annotations and validation. The repair ledger records each finding and whether it was fixed, disclosed or declined, with an evidence path. The historical review is preserved so that a later passing check cannot erase the reason a calculation was changed.

The main numerical change is the withdrawal of Pakistan's complete intake estimate. The Indian benchmark now retains the consumer-unit denominator and uses the EAR as the primary adult scenario. Derived outputs are identified as calculations rather than official observations. Company and policy entries retain dates, exclusions and documentary scope. Regional and nutrition displays keep incompatible measures apart.

The annotation work is confined to the limitations appendix. Collection stopped, subject-country and topic eligibility were reviewed, and an independent fresh-context pass was compared with the earlier labels. That comparison describes agreement between annotation runs. It is not accuracy, human adjudication, public opinion or evidence of consumer adoption.

Final browser evidence must correspond to the actual edition, and publication files require a reviewed allowlist. Rebuilds, source joins and export parity test implementation integrity. They cannot substitute for the source's research design or for a qualified reader's judgment about an interpretation. The report's remaining scientific limits survive a successful software check.

Pakistan still lacks a compatible complete-diet quantity and denominator basis in this model. The consumer evidence lacks a common bilateral instrument and observed adoption baseline. Current domestic sales, verified retail availability, operating capacities and product-specific approvals remain unevenly documented. These are missing research inputs, not empty cells to be filled with a scenario midpoint.

The public-source search has a defined stopping point. Failure to retrieve a source in that search means it was not established by this work, not that it cannot exist. Source ages remain visible, and access dates are not substituted for publication or fieldwork dates. A later researcher can revise a finding with new evidence while preserving this edition's record.

The next data collection should answer a named question

The first task is to establish whether the identified respondent-data routes contain reusable observations with compatible product and outcome definitions. Before analysis, each acquisition should record the file version, licence, sampling frame, language, recruitment, fieldwork dates, exclusions and exact variable meanings. A public repository title is insufficient evidence for any of these details.

A product-availability study would separately define locations, retailers, observation dates and what counts as available. An import-to-order listing should remain distinct from an item physically in stock. Recording package size and price per relevant serving would support an affordability comparison, while a stock record alone would not establish sales.

A small consumer pilot could then test the meal-based comparison proposed in the brief. It should include people with different food expenditures, dietary practices and responsibilities for shopping or cooking. The immediate outcome would be whether the product description and choice task are understood. A purposive pilot would not estimate national demand.

Repeated purchasing and displacement require follow-up observations. A diary or transaction study should record whether the new product replaced meat, dairy, pulses or another food, and whether the change persisted. Those outcomes would make the later forecast question more concrete. Until such evidence exists, the appropriate contribution is a traceable account of current conditions and explicit unanswered questions.

Appendix A Source ledger

Sources cited in this report are listed by stable identifier, title, tier, recorded vintage, access date and public URL where available. The complete source CSV also preserves discovery records, failed retrievals and held sources, with publishers, rights notes and access limitations. An empty public link marks a local research record. Inclusion in the companion ledger is not approval for substantive use.

Source record
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IN-D01: Survey Dataset on Consumer Acceptance of Cultured Meat in India; tier C; vintage Not recorded; accessed 2026-10-11; Open source

IN-D02: Adoption of Cultured Meat in India; tier C; vintage Not recorded; accessed 2026-10-11; Open source

IN-D03: A Survey of Consumer Perceptions of Plant-Based and Clean Meat in the USA, India, and China; tier C; vintage Not recorded; accessed 2026-10-11; Open source

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icmr_nin_dietary_guidelines_2024: Dietary Guidelines for Indians, revised edition 2024; tier A; vintage 2024; accessed 2026-10-13; Open source

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usda_sr_legacy_documentation_2018: USDA National Nutrient Database for Standard Reference Legacy2018 documentation; tier A; vintage 2018; accessed 2026-10-13; Open source
worldbank_adept_food_security_2014: Analyzing Food Security Using Household Survey Data; tier A; vintage 2014; accessed 2026-10-13; Open source

Appendix B Claims ledger summary

The rows below summarize numerical bindings used in the report. The complete claims CSV retains each exact filter, operation, rounding rule and original locator. Repeated quantities can share a dataset but refer to different groups or scenarios, so the filter must be read before comparison. News-framing results are confined to their separate appendix.

Claim record
acquisition_in_d01_sample_n. phase two leads; route id IN-D01; sample_n: 257 source-reported candidate sample; tier derived; Source
acquisition_in_d02_sample_n. phase two leads; route id IN-D02; sample_n: 350 source-reported candidate sample; tier derived; Source
acquisition_in_d03_sample_n. phase two leads; route id IN-D03; sample_n: 1024 source-reported candidate sample; tier derived; Source
acquisition_pk_d01_sample_n. phase two leads; route id PK-D01; sample_n: 649 source-reported candidate sample; tier derived; Source
benchmark_india_animal_coverage_count. benchmark coverage; India; animal-source protein supply; covered: 1 evaluation cells; tier derived; Source
benchmark_india_animal_evaluation_count. benchmark coverage; India; animal-source protein supply; n_evaluated: 6 evaluation cells; tier derived; Source
benchmark_india_animal_forecast_2035. benchmark forecast; India; animal-source protein supply; year 2035; point estimate: 20.01 g/cap/d; tier derived; Source; Source
benchmark_india_animal_lower80_2035. benchmark forecast; India; animal-source protein supply; year 2035; lower interval bound: 16.67 g/cap/d; tier derived; Source; Source

benchmark_india_animal_upper80_2035. benchmark forecast; India; animal-source protein supply; year 2035; upper interval bound: 23.35 g/cap/d; tier derived; Source; Source

benchmark_india_total_coverage_count. benchmark coverage; India; total protein supply; covered: 1 evaluation cells; tier derived; Source

benchmark_india_total_evaluation_count. benchmark coverage; India; total protein supply; n_evaluated: 6 evaluation cells; tier derived; Source

benchmark_india_total_forecast_2035. benchmark forecast; India; total protein supply; year 2035; point estimate: 74.57 g/cap/d; tier derived; Source; Source

benchmark_india_total_lower80_2035. benchmark forecast; India; total protein supply; year 2035; lower interval bound: 68.14 g/cap/d; tier derived; Source; Source

benchmark_india_total_nominal_level. benchmark coverage; India; total protein supply; nominal_level: 80 percent; tier derived; Source

benchmark_india_total_upper80_2035. benchmark forecast; India; total protein supply; year 2035; upper interval bound: 81.00 g/cap/d; tier derived; Source; Source

benchmark_pakistan_animal_coverage_count. benchmark coverage; Pakistan; animal-source protein supply; covered: 6 evaluation cells; tier derived; Source

benchmark_pakistan_animal_evaluation_count. benchmark coverage; Pakistan; animal-source protein supply; n_evaluated: 6 evaluation cells; tier derived; Source

benchmark_pakistan_animal_forecast_2035. benchmark forecast; Pakistan; animal-source protein supply; year 2035; point estimate: 32.55 g/cap/d; tier derived; Source; Source

benchmark_pakistan_animal_lower80_2035. benchmark forecast; Pakistan; animal-source protein supply; year 2035; lower interval bound: 28.50 g/cap/d; tier derived; Source; Source

benchmark_pakistan_animal_upper80_2035. benchmark forecast; Pakistan; animal-source protein supply; year 2035; upper interval bound: 36.60 g/cap/d; tier derived; Source; Source

benchmark_pakistan_total_coverage_count. benchmark coverage; Pakistan; total protein supply; covered: 6 evaluation cells; tier derived; Source

benchmark_pakistan_total_evaluation_count. benchmark coverage; Pakistan; total protein supply; n_evaluated: 6 evaluation cells; tier derived; Source

benchmark_pakistan_total_forecast_2035. benchmark forecast; Pakistan; total protein supply; year 2035; point estimate: 73.39 g/cap/d; tier derived; Source; Source

benchmark_pakistan_total_lower80_2035. benchmark forecast; Pakistan; total protein supply; year 2035; lower interval bound: 65.23 g/cap/d; tier derived; Source; Source

benchmark_pakistan_total_upper80_2035. benchmark forecast; Pakistan; total protein supply; year 2035; upper interval bound: 81.55 g/cap/d; tier derived; Source; Source

fbs_2023_india_milk. meat supply national; country India; item Milk - Excluding Butter; metric food_supply_quantity; year 2023; value: 87.43 kg/capita/year; tier A; Source

fbs_2023_india_poultry. meat supply national; country India; item Poultry Meat; metric food_supply_quantity; year 2023; value: 3.52 kg/capita/year; tier A; Source

fbs_2023_india_pulses. meat supply national; country India; item Pulses; metric food_supply_quantity; year 2023; value: 15.39 kg/capita/year; tier A; Source

fbs_2023_pakistan_milk. meat supply national; country Pakistan; item Milk - Excluding Butter; metric food_supply_quantity; year 2023; value: 129.71 kg/capita/year; tier A; Source

fbs_2023_pakistan_poultry. meat supply national; country Pakistan; item Poultry Meat; metric food_supply_quantity; year 2023; value: 8.72 kg/capita/year; tier A; Source

fbs_2023_pakistan_pulses. meat supply national; country Pakistan; item Pulses; metric food_supply_quantity; year 2023; value: 6.53 kg/capita/year; tier A; Source

hies_chicken_meat_q1. diet composition; country Pakistan; food group chicken_meat; group all_Q1; metric monthly_per_capita_consumption_quantity; region Pakistan; value: 0.17 kg/person/month; tier A; Source
hies_chicken_meat_q5. diet composition; country Pakistan; food group chicken_meat; group all_Q5; metric monthly_per_capita_consumption_quantity; region Pakistan; value: 0.65 kg/person/month; tier A; Source
hies_household_size_q1. household size context; quintile Q1; mean_household_size: 7.56 persons/household; tier A; Source
hies_household_size_q5. household size context; quintile Q5; mean_household_size: 4.65 persons/household; tier A; Source
india_cu_rural_mpce_all_india. protein intake consumer unit; group rural_MPCE_All-India; intake_g: 66.6 g/consumer_unit/day; tier A; Source
india_cu_urban_mpce_all_india. protein intake consumer unit; group urban_MPCE_All-India; intake_g: 69.9 g/consumer_unit/day; tier A; Source
inventory_altprotein_companies_rows. inventory counts; dataset id altprotein_companies; row_count: 16 rows; tier derived; Source
inventory_altprotein_company_exclusions_rows. inventory counts; dataset id altprotein_company_exclusions; row_count: 2 rows; tier derived; Source
inventory_altprotein_market_rows. inventory counts; dataset id altprotein_market; row_count: 19 rows; tier derived; Source
inventory_altprotein_policy_rows. inventory counts; dataset id altprotein_policy; row_count: 16 rows; tier derived; Source
inventory_altprotein_retail_offers_rows. inventory counts; dataset id altprotein_retail_offers; row_count: 1 rows; tier derived; Source
inventory_altprotein_sources_rows. inventory counts; dataset id altprotein_sources; row_count: 111 rows; tier derived; Source
inventory_benchmark_annual_rows. inventory counts; dataset id benchmark_annual; row_count: 56 rows; tier derived; Source
inventory_benchmark_coverage_rows. inventory counts; dataset id benchmark_coverage; row_count: 4 rows; tier derived; Source
inventory_benchmark_forecast_rows. inventory counts; dataset id benchmark_forecast; row_count: 48 rows; tier derived; Source
inventory_boundary_licenses_rows. inventory counts; dataset id boundary_licenses; row_count: 2 rows; tier derived; Source
inventory_claims_rows. inventory counts; dataset id claims; row_count: 642 rows; tier derived; Source
inventory_cultural_context_rows. inventory counts; dataset id cultural_context; row_count: 20 rows; tier derived; Source
inventory_deficiency_population_rows. inventory counts; dataset id deficiency_population; row_count: 28 rows; tier derived; Source
inventory_diet_composition_rows. inventory counts; dataset id diet_composition; row_count: 814 rows; tier derived; Source
inventory_household_size_context_rows. inventory counts; dataset id household_size_context; row_count: 6 rows; tier derived; Source
inventory_meat_regional_rows. inventory counts; dataset id meat_regional; row_count: 2393 rows; tier derived; Source
inventory_meat_supply_national_rows. inventory counts; dataset id meat_supply_national; row_count: 3221 rows; tier derived; Source
inventory_package_metrics_rows. inventory counts; dataset id package_metrics; row_count: 61 rows; tier derived; Source
inventory_packages_rows. inventory counts; dataset id packages; row_count: 12 rows; tier derived; Source
inventory_phase_two_leads_rows. inventory counts; dataset id phase_two_leads; row_count: 43 rows; tier derived; Source
inventory_protein_gap_group_summary_rows. inventory counts; dataset id protein_gap_group_summary; row_count: 18 rows; tier derived; Source
inventory_protein_gap_rows. inventory counts; dataset id protein_gap; row_count: 192 rows; tier derived; Source
inventory_protein_intake_consumer_unit_rows. inventory counts; dataset id protein_intake_consumer_unit; row_count: 26 rows; tier derived; Source
inventory_protein_intake_modelled_rows. inventory counts; dataset id protein_intake_modelled; row_count: 6 rows; tier derived; Source

inventory_protein_intake_observed_rows. inventory counts; dataset id protein_intake_observed; row_count: 26 rows; tier derived; Source

inventory_protein_model_food_coverage_rows. inventory counts; dataset id protein_model_food_coverage; row_count: 38 rows; tier derived; Source

inventory_protein_model_method_ranges_rows. inventory counts; dataset id protein_model_method_ranges; row_count: 12 rows; tier derived; Source

inventory_protein_model_physical_subtotals_rows. inventory counts; dataset id protein_model_physical_subtotals; row_count: 156 rows; tier derived; Source

inventory_protein_model_sensitivity_rows. inventory counts; dataset id protein_model_sensitivity; row_count: 26 rows; tier derived; Source

inventory_raw_manifest_rows. inventory counts; dataset id raw_manifest; row_count: 1974 rows; tier derived; Source

inventory_reference_requirements_rows. inventory counts; dataset id reference_requirements; row_count: 12 rows; tier derived; Source

inventory_region_boundary_crosswalk_rows. inventory counts; dataset id region_boundary_crosswalk; row_count: 43 rows; tier derived; Source

inventory_review_status_rows. inventory counts; dataset id review_status; row_count: 29 rows; tier derived; Source

inventory_sources_rows. inventory counts; dataset id sources; row_count: 908 rows; tier derived; Source

inventory_survey_findings_rows. inventory counts; dataset id survey_findings; row_count: 29 rows; tier derived; Source

inventory_survey_studies_rows. inventory counts; dataset id survey_studies; row_count: 9 rows; tier derived; Source

nutrition_2024_india_stunting. deficiency population; country India; indicator under5_stunting; year 2024; value: 32.9 percent; tier A; Source

nutrition_2024_india_unaffordable. deficiency population; country India; indicator unable_to_afford_healthy_diet; year 2024; value: 40.4 percent; tier A; Source

nutrition_2024_pakistan_stunting. deficiency population; country Pakistan; indicator under5_stunting; year 2024; value: 33.6 percent; tier A; Source

nutrition_2024_pakistan_unaffordable. deficiency population; country Pakistan; indicator unable_to_afford_healthy_diet; year 2024; value: 60.3 percent; tier A; Source

oecd_2034_india_poultry_meat_oecd_human_consumption_projection. meat supply national; country India; item Poultry meat; metric oecd_human_consumption_projection; year 2034; value: 2.30 kg/person/year (retail weight); tier A; Source

oecd_2034_pakistan_poultry_meat_oecd_human_consumption_projection. meat supply national; country Pakistan; item Poultry meat; metric oecd_human_consumption_projection; year 2034; value: 5.70 kg/person/year (retail weight); tier A; Source

package_p06_access_routes. package metrics; metric access_routes; package id P06; value: 43 records; tier derived; Source

package_p06_contextual_routes. package metrics; metric contextual_routes; package id P06; value: 13 records; tier derived; Source

package_p07_finding_records. package metrics; metric finding_records; package id P07; value: 29 findings; tier derived; Source

package_p07_study_records. package metrics; metric study_records; package id P07; value: 9 studies; tier derived; Source

package_p08_annual_observations. package metrics; metric annual_observations; package id P08; value: 56 observations; tier derived; Source

package_p08_selected_forecasts. package metrics; metric selected_forecasts; package id P08; value: 48 forecasts; tier derived; Source

package_p09_human_verification_pending. package metrics; metric human_verification_pending; package id P09; value: 95 records; tier derived; Source

package_p09_source_records. package metrics; metric source_records; package id P09; value: 95 records; tier derived; Source

package_p09_sources_held. package metrics; metric sources_held; package id P09; value: 22 records; tier derived; Source

package_p09_sources_not_retrieved. package metrics; metric sources_not_retrieved; package id P09; value: 10 records; tier derived; Source

package_p09_sources_verified_with_note. package metrics; metric sources_verified_with_note; package id P09; value: 63 records; tier derived; Source

pakistan_model_imputed_food_count. protein model sensitivity; scenario id edible_flesh_us_large; imputed_food_count: 0 foods; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_model_n_converted_foods. protein model sensitivity; scenario id edible_flesh_us_large; n_converted_foods: 27 foods; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_model_n_table23_foods. protein model sensitivity; scenario id edible_flesh_us_large; n_table23_foods: 38 foods; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_partial_all_all_five_animal_foods_subtotal_high. protein model method ranges; group all_all; scope five_animal_foods_subtotal; high method endpoint: 4.64 g/person/day; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_partial_all_all_five_animal_foods_subtotal_low. protein model method ranges; group all_all; scope five_animal_foods_subtotal; low method endpoint: 3.30 g/person/day; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_partial_all_all_matched_basket_subtotal_high. protein model method ranges; group all_all; scope matched_basket_subtotal; high method endpoint: 40.61 g/person/day; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

pakistan_partial_all_all_matched_basket_subtotal_low. protein model method ranges; group all_all; scope matched_basket_subtotal; low method endpoint: 39.27 g/person/day; tier derived; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source; Source

requirement_india_ear_60kg. reference requirements; benchmark id india_EAR_60kg; requirement_g_day: 39.60 g/day; tier derived; Source; Source

requirement_india_ear_60kg_coefficient. reference requirements; benchmark id india_EAR_60kg; protein_g_per_kg_day: 0.66 g/kg/day; tier derived; Source; Source

requirement_india_ear_60kg_weight. reference requirements; benchmark id india_EAR_60kg; reference_body_weight_kg: 60 kg; tier derived; Source; Source

requirement_india_rda_60kg. reference requirements; benchmark id india_RDA_60kg; requirement_g_day: 49.80 g/day; tier derived; Source; Source

requirement_india_rda_60kg_coefficient. reference requirements; benchmark id india_RDA_60kg; protein_g_per_kg_day: 0.83 g/kg/day; tier derived; Source; Source

study_s01_sample_n. survey studies; study id S01; sample_size: 1025 reported study respondents; tier derived; Source

study_s02_sample_n. survey studies; study id S02; sample_size: 1666 reported study respondents; tier derived; Source

study_s03_sample_n. survey studies; study id S03; sample_size: 4488 reported study respondents; tier derived; Source

study_s04_sample_n. survey studies; study id S04; sample_size: 1169 reported study respondents; tier derived; Source

study_s05_sample_n. survey studies; study id S05; sample_size: 2208 reported study respondents; tier derived; Source

study_s06_sample_n. survey studies; study id S06; sample_size: 102 reported study respondents; tier derived; Source

study_s07_sample_n. survey studies; study id S07; sample_size: 78816 reported study respondents; tier derived; Source

study_s08_sample_n. survey studies; study id S08; sample_size: 5971 reported study respondents; tier derived; Source

study_s09_sample_n. survey studies; study id S09; sample_size: 2000 reported study respondents; tier derived; Source

survey_r01_sample_n. survey findings; record id R01; sample_n: 8305 source-reported subgroup respondents; tier derived; Source

survey_r01_value_percent. survey findings; record id R01; value_percent: 31.0 percent; tier derived; Source

survey_r02_sample_n. survey findings; record id R02; sample_n: 7706 source-reported subgroup respondents; tier derived; Source
survey_r02_value_percent. survey findings; record id R02; value_percent: 50.0 percent; tier derived; Source
survey_r03_sample_n. survey findings; record id R03; sample_n: 7474 source-reported subgroup respondents; tier derived; Source
survey_r03_value_percent. survey findings; record id R03; value_percent: 37.0 percent; tier derived; Source
survey_r04_sample_n. survey findings; record id R04; sample_n: 3507 source-reported subgroup respondents; tier derived; Source
survey_r04_value_percent. survey findings; record id R04; value_percent: 35.0 percent; tier derived; Source
survey_r05_sample_n. survey findings; record id R05; sample_n: 10176 source-reported subgroup respondents; tier derived; Source
survey_r05_value_percent. survey findings; record id R05; value_percent: 32.0 percent; tier derived; Source
survey_r06_sample_n. survey findings; record id R06; sample_n: 1942 source-reported subgroup respondents; tier derived; Source
survey_r06_value_percent. survey findings; record id R06; value_percent: 9.0 percent; tier derived; Source
survey_r14_sample_n. survey findings; record id R14; sample_n: 1666 source-reported subgroup respondents; tier derived; Source
survey_r14_value_percent. survey findings; record id R14; value_percent: 74.0 percent; tier derived; Source
survey_r15_sample_n. survey findings; record id R15; sample_n: 1666 source-reported subgroup respondents; tier derived; Source
survey_r15_value_percent. survey findings; record id R15; value_percent: 12.0 percent; tier derived; Source
survey_r16_sample_n. survey findings; record id R16; sample_n: 1666 source-reported subgroup respondents; tier derived; Source
survey_r16_value_percent. survey findings; record id R16; value_percent: 9.0 percent; tier derived; Source
survey_r17_sample_n. survey findings; record id R17; sample_n: 1666 source-reported subgroup respondents; tier derived; Source
survey_r17_value_percent. survey findings; record id R17; value_percent: 5.0 percent; tier derived; Source
survey_r23_sample_n. survey findings; record id R23; sample_n: 1169 source-reported subgroup respondents; tier derived; Source
survey_r23_value_percent. survey findings; record id R23; value_percent: 23.0 percent; tier derived; Source
survey_r24_sample_n. survey findings; record id R24; sample_n: 758 source-reported subgroup respondents; tier derived; Source
survey_r24_value_percent. survey findings; record id R24; value_percent: 24.1 percent; tier derived; Source
survey_r25_sample_n. survey findings; record id R25; sample_n: 1450 source-reported subgroup respondents; tier derived; Source
survey_r25_value_percent. survey findings; record id R25; value_percent: 7.8 percent; tier derived; Source

Appendix C Review resolutions

Each finding retains its recorded state. These descriptions are concise; the review CSV preserves stable identifiers and the detailed repair account. Open final-output requirements remain open until their evidence exists.

Review finding and resolution
Shape checker accepts corrupted data. Recorded status: fixed. Recompute claims and test deliberate corruptions; historical shape checks remain dated.
Pakistan animal-food protein uses basket-wide imputation. Recorded status: fixed. Withdraw the pooled-basket total and use named physical conversions for partial foods.

One Pakistan model value hides method sensitivity. Recorded status: fixed. Bind partial-food range endpoints to named methods and coverage; withdraw the old complete-total range.

Independent second framing annotation absent. Recorded status: fixed. Compare a fresh-context annotation with the earlier labels and disclose uncertainty and unknown earlier model identity.

Publisher market mistaken for subject country. Recorded status: fixed. Keep publisher and subject geography separate; reconcile exclusions and holds; confine results to the report appendix.

UnCrave closure absent. Recorded status: fixed. Retain the reported UnCrave closure as a dated event with an unknown exact cessation date.

FSSAI plant-based rules absent. Recorded status: fixed. Add the vegan-food rules and dairy-analogue consultation with their distinct legal status.

Adult RDA compared with all-age means. Recorded status: fixed. Use published consumer-unit intake with a primary EAR scenario and an explicit denominator caveat.

Pakistan rural/urban ranking caused by imputation. Recorded status: fixed. Remove rural and urban model rows and rankings from active projections while preserving historical artifacts.

Official and experimental protein plotted as comparable. Recorded status: fixed. Place incompatible measures in separate panels and retain their source definitions.

Unweighted group share resembles population prevalence. Recorded status: fixed. Remove the unweighted fraction of groups below the reference; it was not a population rate.

Derived rows inherit input tiers; blanks remain. Recorded status: fixed. Tag analyst calculations as derived and retain input-source tiers separately.

Apply the revised freshness cutoff and integrate the saved parliamentary programme update. Recorded status: fixed. Apply the revised freshness cutoff and integrate the saved parliamentary programme update; retain the old rule as historical.

Uneven food/feed inclusion; Pakistan coverage thin. Recorded status: fixed. Apply the human-food versus feed rule consistently and retain explicit exclusions.

Browser QA predates generated page. Recorded status: fixed. Require browser receipts for the actual edition bytes; old screenshots do not validate this page.

Self-authored blocked status counts as success. Recorded status: fixed. Fail missing evidence and skipped required checks; self-authored completion or blocked states cannot produce a pass.

Broad Pakistan rule-absence claim is stale. Recorded status: fixed. Correct the broad Pakistan rule-absence statement while distinguishing the general provincial provision from cultivated-specific approval.

Correct the revised HIES quantity denominator to per capita.. Recorded status: fixed. Correct the revised HIES quantity denominator to per capita.

Name the Pakistan demographic and health survey by its actual fieldwork period.. Recorded status: fixed. Name the Pakistan demographic and health survey by its actual fieldwork period.

Sample/date metadata and badges hardcoded. Recorded status: fixed. Read sample and fieldwork metadata from source rows, retain unknowns and populate badges from actual row and source tiers.

Pakistan policy metadata and filing date missing. Recorded status: fixed. Add Pakistan policy date, status and freshness; retain the Biokraft filing date separately from its list date.

Historical coverage receipt disagrees with current corpus. Recorded status: fixed. Reconcile annotation coverage to every original record, including unresolved extent holds.

Large sentiment JSON stored with scripts. Recorded status: fixed. Move large annotation inputs into ignored local storage with hashes, compatibility references and an undo record.

Environment, caches and PNGs clutter versioning. Recorded status: fixed. Exclude environments, caches and screenshots from versioning without deleting local evidence.

Dashboard lacked Git history. Recorded status: fixed. Create a local baseline and a repair branch for the dashboard; no remote is configured.
Governor review artifacts require preservation. Recorded status: disclosed. Preserve internal review history untouched and exclude it from the publication export.
Historical retrieval times may use filesystem mtime. Recorded status: fixed. Require explicit retrieval timestamps for new evidence and preserve original identity and dates on routine reuse.
Calibration tests rebuild frozen outputs in place. Recorded status: fixed. Run calibration tests in temporary replicas and verify that frozen research outputs remain unchanged.
Public export could include copyrighted article bodies. Recorded status: fixed. Require an explicit publication allowlist with reviewed file hashes and rights.

Appendix D Data inventory

The inventory distinguishes table rows from observations, people and independent sources. The raw-manifest count refers to preserved local inputs, which are not redistributed as a whole. Dataset versions and source locators accompany the CSVs. The original package inventory remains a dated historical checkpoint; these counts describe the assembled edition/report data.

Table or record set	Rows	Unit of record and boundary
altprotein companies	16	Dated company/activity records; inclusion does not verify current sales or operating capacity.
altprotein company exclusions	2	Candidates excluded by the human-food scope rule; retained for an auditable boundary.
altprotein market	19	Source-specific market/context quantities; geography, product definition and units are not interchangeable.
altprotein policy	16	Dated rules, drafts and programme events; a policy mention is not product authorization.
altprotein retail offers	1	Observed listing evidence; an offer does not establish local stock, sales or demand.
altprotein sources	111	Feedstock, ingredient and capacity context records; production potential is not alternative-protein adoption.
cultural context	20	Source-bounded cultural observations and survey context; affiliation alone does not identify purchasing behaviour.
deficiency population	28	Country/period/proxy indicators with separate denominators; proxies do not identify protein-deficient people.
diet composition	814	Food/group/period nutrient-share, quantity or expenditure records; units and household bases remain separate.

meat regional	2393	Region/indicator/period records including production and reported food frequency; these cannot be pooled as intake.
meat supply national	3221	Country/item/measure/year supply and projection records; availability and forecasts differ from consumed intake.
protein gap	192	Group/reference scenarios; comparisons are not population deficiency prevalence, and incompatible cases remain withheld.
protein gap group summary	18	Residence/reference summaries of tabulated groups; group counts are not population-weighted rates.
protein intake consumer unit	26	Published Indian group/sector estimates per consumer unit; the denominator is an energy-equivalence scale.
protein intake modelled	6	Withdrawn complete Pakistan group estimates; intake values remain blank.
protein intake observed	26	Published Indian group/sector estimates per person; preserve their distinct denominator.
protein model food coverage	38	Food conversion coverage records, including unresolved and excluded inputs; a row is not a complete diet.
protein model method ranges	12	Group/subtotal endpoints across named methods; conditional ranges are not confidence intervals.
protein model physical subtotals	156	Method/group combinations for incomplete physical-food subtotals; total dietary intake is not inferred.
protein model sensitivity	26	Named conversion-method scenarios; no scenario establishes an observed national intake.
region boundary crosswalk	43	Source-region to boundary matches and holds; geometries do not validate the attached indicator.
packages	12	Research-package coverage records; coverage is separate from scholarly approval.
package metrics	61	Package-specific scope and historical artifact counts; these are not independent empirical findings.
benchmark annual	56	Supply observations by country, protein series and year, used by the benchmark; not alternative-protein consumption.
benchmark forecast	48	Point and interval outputs by country, series and target year; exploratory supply forecasts, not adoption forecasts.
benchmark coverage	4	Series-specific evaluation metrics and denominators; small evaluation samples limit calibration claims.

survey studies	9	Study descriptions with separate samples and recruitment; respondent totals must not be pooled.
survey findings	29	Heterogeneous reported outcomes, including held findings; only explicitly admitted outcomes enter the synthesis.
phase two leads	43	Access-route metadata; source-reported samples do not mean respondent files were acquired.
reference requirements	12	Country/reference/weight scenarios; published coefficients and assumed body weights are distinct inputs.
household size context	6	Pakistani group-level household-size context; these values do not convert quantities into protein-equivalent people.
review status	29	Review findings and recorded resolutions; software repair does not establish human scholarly approval.
boundary licenses	2	Boundary-source and distribution attribution records; rights statements are asset-specific.
sources	908	Registered source identities; records need not be independent studies or currently accessible links.
raw manifest	1974	Preserved raw-file manifest entries; retrieval and file counts do not establish research validity or redistribution rights.
claims	642	Numeric display bindings to selected source rows, including appendix-only claims; repeated displays can share one claim.

Appendix E Glossary

The key gives general categories; retained records can carry a more conservative historical grade. The survey extraction keeps B for sampled papers and C for the GFI commissioned market survey. These grades are preserved for provenance and do not establish a new quality review.

Term	Meaning here
SOFI	The State of Food Security and Nutrition in the World: the interagency report series supplying the affordability estimates.
JME	Joint Child Malnutrition Estimates, produced by UNICEF, WHO and the World Bank.
MUIS	Majlis Ugama Islam Singapura, Singapore's Islamic Religious Council.
FSSAI	Food Safety and Standards Authority of India.
BioE3	Biotechnology for Economy, Environment and Employment, India's biotechnology policy programme.

FBS	Food balance sheets: national accounting estimates of food available for human consumption.
HIES	Household Integrated Economic Survey: the Pakistani household survey and its stated quantity or expenditure measures.
HCES	Household Consumption Expenditure Survey: the Indian household survey used for published nutrient estimates.
MPCE and expenditure fractiles	Monthly per-capita consumption expenditure. Fractiles rank the population from lower to higher expenditure, separately within rural and urban sectors; they do not measure protein deficiency.
NFHS	National Family Health Survey: respondent and household health measures, including reported food frequency.
EAR	Estimated Average Requirement: a requirement reference, not an observed individual intake.
RDA	Recommended Dietary Allowance: a distinct reference retained here as a labelled sensitivity.
Consumer unit	An energy-based equivalence scale, not a protein-equivalent person.
Tier	A source category whose use remains bounded by method, date, denominator and claim scope.
Tier A	National statistics offices, ministries, FAO/WHO/World Bank/OECD datasets, and peer-reviewed papers with a stated sample.
Tier B	GFI, USDA FAS GAIN, UN agency reports and reputable trackers under the project classification.
Tier C	Market-research summaries, news and company press releases; a recorded claim is not independent confirmation.
Tier D	Social media, Google Trends and LLM output. These cannot supply headline evidence; annotation results are confined to the report appendix.
Derived	An analyst calculation or model output; the source tier is retained separately.
Acceptance	A study-specific response that must not be substituted for observed purchase or habitual use.
Adoption	An outcome requiring a stated product, population, behaviour, denominator and period.
Halal certification	A scoped institutional decision for a product and process, distinct from general religious conditions and food authorization.

Appendix F News framing remains annotation evidence

Collection stopped. The bounded review reconciles 194 original records to 28 admissible items, 164 excluded items and 2 additional extent holds. The admissible set contains 27 India-focused items and 1 Pakistan-focused item. That composition does not support a country comparison or an estimate of public sentiment.

The second annotation ran without the earlier labels. On the admissible primary set, the runs agree on 25 of 28 items. Cohen’s kappa is 0.798561, with exact arithmetic 111 divided by 139. The comparison is conditioned on the second run’s eligibility decisions and the source-access holds.

A sensitivity retaining the second run’s held extents has 26 agreements among 30 items, with kappa 0.759036. Its inclusion does not resolve the extent holds. The primary set includes 6 high-uncertainty second-run decisions. Neither statistic measures accuracy or human consensus.

```
Second-run runtime model: gpt-6-astra
Earlier-run runtime model: not established
Human adjudication: pending
```

The method record preserves the prompt, immutable inputs, individual labels, exclusions and disagreements locally. Runtime identity does not establish independent model families, and the unknown earlier identity is not filled by inference. Only bounded metadata and derived results are prepared for the report; article bodies are excluded from publication. No framing statistic appears in the edition’s content or downloads.